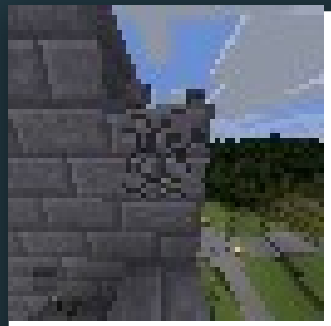
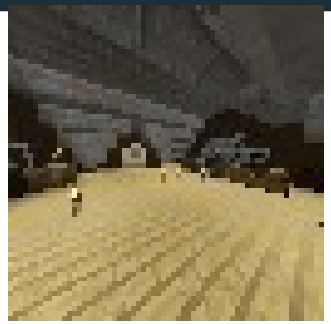
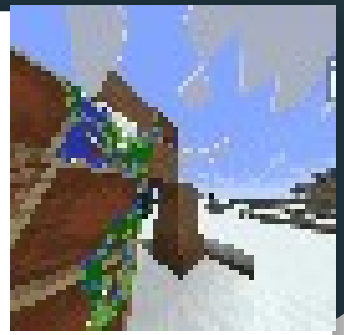




FL26 - Agent 37
AI & ML@SJSU



Project Lead: Ben Pham
Confirmed Project Member: Vikram Arunkumar



Problem Description / Goal

- Build an agent that learns to act in Minecraft from pixels and human demonstrations
- Start with BASALT-style tasks where success is judged by people, not only a hand-written reward
- Reproduce a behavioral-cloning baseline, then improve sample efficiency and task completion
- Stretch goal: compare imitation learning with reinforcement-learning fine-tuning
- Deliverable: a reproducible training and evaluation pipeline

Why Minecraft makes a strong AI benchmark

- Rich, visual, open-ended world: one environment supports navigation, crafting, resource collection, and building
- Long-horizon tasks require memory, exploration, and hierarchical planning
- MineRL combines Gym-compatible environments with datasets of human gameplay
- The gap between a valid action and a useful outcome makes evaluation scientifically interesting
- A shared benchmark lets students compare approaches on the same tasks

Datasets & Environment Interface

- Human demonstrations: MineRL gameplay trajectories for imitation learning
- Task suite: BASALT environments such as FindCave, MakeWaterfall, and BuildVillageHouse
- Observation: first-person RGB image; the v0.4 docs specify 360 × 640 × 3
- Action: structured dictionary for movement, camera, inventory, hotbar, attack, and use
- Project data: recorded trajectories, actions, task outcomes, and experiment metadata

High Level Plan

- Install and validate a MineRL environment
- Inspect demonstrations and collect random-agent trajectories
- Train a behavioral-cloning baseline: POV frames → actions
- Evaluate on held-out seeds with task and human-judgment rubrics
- Iterate with temporal context, action abstraction, and RL fine-tuning

Related Work / Research

- MineRL: A Large-Scale Dataset of Minecraft Demonstrations
- MineRL 2019 and 2020 competitions: sample-efficient RL with human priors
- BASALT: human-judged tasks and learning from demonstrations or preferences
- Behavioral-cloning and VPT-style baselines provide practical starting points
- https://minerl.readthedocs.io/en/latest/
- https://github.com/minerllabs/minerl

Project Evaluation Metrics

- Task completion rate on held-out seeds
- Human preference or rubric score for BASALT-style tasks
- Sample efficiency: performance versus environment interactions
- Action validity, episode survival, and time-to-completion
- Reproducibility: setup time, deterministic evaluation, and documented configs

Tech Stack - Desired Member Experience

- Python + MineRL + Gym-compatible environment interface
  - PyTorch for policy learning; NumPy/OpenCV for trajectory preparation
- Jupyter plus TensorBoard or Weights & Biases for experiment tracking
  - Git/GitHub plus Docker or conda for reproducible environments
- Members leave with experience in RL, imitation learning, embodied AI, and evaluation

Roles Needed

- General project contributor
- Environment & data pipeline
- Imitation learning / behavioral cloning
- RL, metrics, and human evaluation
- Experiment infrastructure and reproducibility

Let's Get Connected

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